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RESEARCH-ARTICLE

DistillRecDial: A Knowledge-Distilled Dataset Capturing User Diversity in Conversational Recommendation

ALESSANDRO FRANCESCO MARIA MARTINA, University of Bari Aldo Moro, Bari, BA, Italy

ALESSANDRO PETRUZZELLI, University of Bari Aldo Moro, Bari, BA, Italy

CATALDO MUSTO, University of Bari Aldo Moro, Bari, BA, Italy

MARCO DE GEMMIS, University of Bari Aldo Moro, Bari, BA, Italy

PASQUALE LOPS, University of Bari Aldo Moro, Bari, BA, Italy

GIOVANNI SEMERARO, University of Bari Aldo Moro, Bari, BA, Italy

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Alessandro Francesco Maria
Martina
University of Bari Aldo Moro
Bari, Italy
University of Pisa
Pisa, Italy
alessandro.martina@uniba.it

Marco de Gemmis
University of Bari Aldo Moro
Bari, Italy
marco.degemmis@uniba.it

Alessandro Petruzzelli
University of Bari Aldo Moro
Bari, Italy
alessandro.petruzzelli@uniba.it

Pasquale Lops
University of Bari Aldo Moro
Bari, Italy
pasquale.lops@uniba.it

Cataldo Musto
University of Bari Aldo Moro
Bari, Italy
cataldo.musto@uniba.it

Giovanni Semeraro
University of Bari Aldo Moro
Bari, Italy
giovanni.semeraro@uniba.it

Abstract

Conversational Recommender Systems (CRSs) facilitate item discovery through multi-turn dialogues that elicit user preferences via natural language interaction. This field has gained significant attention following advancements in Natural Language Processing (NLP) enabled by Large Language Models (LLMs). However, current CRS research remains constrained by datasets with fundamental limitations. Human-generated datasets suffer from inconsistent dialogue quality, limited domain expertise, and insufficient scale for real-world application, while synthetic datasets created with proprietary LLMs ignore the diversity of real-world user behavior and present significant barriers to accessibility and reproducibility. The development of effective CRSs depends critically on addressing these deficiencies.

To this end, we present DISTILLREC DIAL, a novel conversational recommendation dataset generated through a knowledge distillation pipeline that leverages smaller, more accessible open LLMs. Crucially, DISTILLREC DIAL simulates a range of user types with varying intentions, preference expression styles, and initiative levels, capturing behavioral diversity that is largely absent from prior work. Human evaluation demonstrates that our dataset significantly outperforms widely adopted CRS datasets in dialogue coherence and domain-specific expertise, indicating its potential to advance the development of more realistic and effective conversational recommender systems.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Human-centered computing** → *User models*.

Keywords

Conversational Recommendation, Datasets, Large Language Models, Large Multimodal Models, Symbiotic AI

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1 Introduction

Conversational Recommender Systems (CRSs) are software systems that assist users in decision-making by providing personalized recommendations through multi-turn interactions [22]. Conversational interaction has become increasingly prominent in recent years, thanks to advancements in the field of Natural Language Processing, and CRSs show great potential in enhancing user satisfaction, trust, and engagement in multiple domains [1, 13].

Since the emergence of end-to-end approaches [29] in CRS development, researchers have primarily focused on building models trained on pre-annotated conversational recommendation datasets [8, 37, 45, 49, 50]. Consequently, the construction of high-quality conversational recommendation datasets has emerged as a critical research priority. Existing conversational recommendation datasets are typically created through one of two approaches: either by capturing interactions between crowdworkers who roleplay as recommendation seekers and recommenders [16, 24, 29], or by employing synthetic generation techniques using powerful proprietary Large Language Models (LLMs) [4, 38] such as GPT-3.5 [17, 28, 33].

Human-generated conversational recommendation datasets often **lack coherence and realism, as crowdworkers do not engage with genuine preferences and intentions** [15, 28]. Furthermore, these participants often produce suboptimal recommendations with inadequate explanations [34], reflecting their limited expertise in the specific recommendation domains [15]. This results in shallow conversations and suboptimal recommendations. Moreover, such datasets are usually small and fail to capture the



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diversity and complexity of user behavior [13]. To highlight these limitations, Table 1 shows a dialogue sampled from ReDial, the de facto standard dataset for end-to-end CRSs

Table 1: Real example of a dialogue from the ReDial dataset.

Recommender:	Hi
Seeker:	Hello, Anything you have in mind?
Seeker:	I just watched a movie called John Wick: Chapter 2 – it was a pretty amazing action film.
Recommender:	I recommend animated movies
Seeker:	Have you see it?
Seeker:	Okay.
Recommender:	Boruto: Naruto the Movie and Dragon Ball Z: Lord Slug

Recent efforts to generate synthetic datasets using LLMs show promise in overcoming some of the issues above [17, 28, 31]. However, a crucial shortcoming remains unaddressed: **the absence of explicit user modeling in terms of intentions, preferences, and behavioral stereotypes**. Indeed, in real-world CRS deployments, user behavior varies significantly; some users provide detailed preferences early, others require more guidance, and different users may respond better to different initiative strategies [5, 46]. Yet, current datasets either ignore these variations entirely or treat users as homogenous entities with static needs. As a result, models trained on such data struggle to generalize to diverse user types, limiting their real-world applicability and user trustworthiness [21].

Moreover, existing works in the area rely heavily on access to proprietary LLMs, which restricts their use to researchers with the necessary resources and access privileges. This dependency results in non-scalable and non-extensible solutions, particularly problematic as many synthetic datasets are built with versions of GPT-3.5 that are no longer publicly available. Additionally, these datasets are **rarely integrated into widely adopted benchmarking frameworks** like CRSLab [48], limiting their reproducibility and accessibility within the broader research community.

To address these gaps, we introduce DISTILLREC DIAL, a novel conversational recommendation dataset that explicitly incorporates diverse user personas and interaction patterns. Unlike prior work, DISTILLREC DIAL models user heterogeneity in terms of preferences, goals, and dialogic behavior, essential dimensions for realistic and effective CRS training and evaluation. We generate DISTILLREC DIAL through a Large-to-Small Language Model Distillation pipeline [19, 41], enabling high-quality dialogue synthesis without relying on closed, resource-intensive LLMs. To ensure both quality and usability, we conduct comprehensive evaluations, including human annotations and automated assessments, which demonstrate that DISTILLREC DIAL consistently outperforms existing datasets in terms of dialogue quality and diversity.

Finally, to encourage reproducibility and adoption, we release the DISTILLREC DIAL dataset¹ along with code integrated into the CRSLab framework², enabling both seamless use of the dataset and reproduction of experiments on representative CRS models.

¹<https://huggingface.co/datasets/swap-uniba/DistillRecDial>

²<https://github.com/swapUniba/CRSLab-DistillRecDial>

2 Related Works

2.1 Datasets for Conversational RecSys

CRSs have emerged as a prominent research focus in recent years, driven by significant advancements in NLP and their distinctive capacity to deliver personalized recommendations through natural language interactions.

The development of CRSs can be classified into two primary categories: attribute-based [12, 35] and end-to-end systems [8, 29]. Attribute-based approaches aim to identify user preferences and generate appropriate recommendations within a constrained number of conversational turns, utilizing pre-defined actions and template-based response generation [10]. Despite demonstrating initial efficacy, these systems necessitate a knowledge engineering phase that inherently restricts their adaptability. Consequently, research has shifted towards end-to-end CRSs - systems founded on deep neural networks that inherently require substantial training dialogue corpora.

Within the domain of end-to-end CRSs, researchers have introduced various human-generated datasets. The foundational work of ReDial [29] presented the first deep neural network-based CRS alongside a crowdsourced dataset featuring movie recommendations integrated with casual conversation. Subsequently, GoRecDial [24] offered a goal-oriented recommendation dataset anchored in authentic user movie preferences. More recently, INSPIRED [16] introduced a human-annotated recommendation dataset with systematically categorized recommendation strategies derived from social science theories. The principal limitation across these contributions stems from their lack of scalability and frequent issues regarding dialogue quality [13].

To overcome these constraints, multiple studies have proposed the generation of synthetic datasets for conversational recommendation. In [17] the authors extracted single-turn recommendation dialogues from Reddit and transformed them into conversational recommendation datasets; however, the practical utility of this approach was compromised by the suboptimal quality of the scraped dialogues [28]. In AUGUST [33], the authors generated recommendation dialogues by converting user-item interactions into conversational formats using a data-to-text generation model. Nevertheless, since this model was trained on existing recommendation dialogues, the generated content retained the inherent limitations of crowdsourced datasets.

The use of LLMs for conversational recommendation dataset generation has been explored in works such as PEARL [28] and LLM-REDIAL [31], which highlight the potential of GPT-3.5 as a user simulator trained on real-world recommendation datasets to generate synthetic conversational data. These approaches demonstrate the potential of large-scale generative models for CRS data creation, yielding outputs that are both coherent and scalable. However, they face significant drawbacks. Most notably, they rely on closed-source APIs that are not publicly reproducible, introducing barriers to transparency, reproducibility, and further development. Additionally, they overlook the heterogeneity of real-world users, modeling users as monolithic entities rather than as diverse individuals with varying goals, preferences, and interaction styles.

Differently from existing LLM-generated datasets, we explicitly encode user variation by modeling distinct user stereotypes, thereby

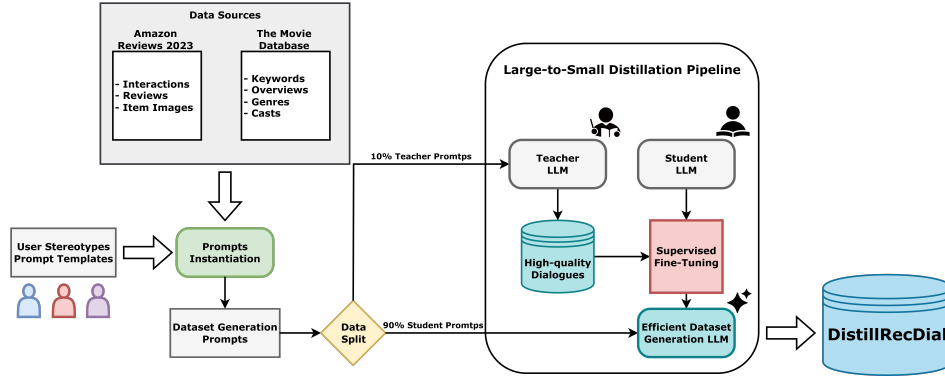


Figure 1: Architecture of the proposed pipeline.

reflecting realistic diversity in conversational behavior. Moreover, we leverage a distillation-based pipeline that utilizes openly available small LLMs guided by large model-generated exemplars to synthesize high-quality dialogues.

2.2 Dialogue Datasets Generation with LLMs

The rise of LLMs has sparked widespread interest in creating dialogue datasets through automated methods.

In [7], the authors utilized an LLM to construct a corpus of dialogues specifically designed for the classification tasks of emotion, speech act, and intent. In [47] the researchers implemented a fine-tuned LLM to develop a specialized dataset for emotional support conversations.

In [26] the authors established a human-AI collaborative framework that leveraged GPT-3 to create dialogue data aimed at training conversational agents to respond appropriately to problematic content in accordance with established social norms. In [27] researchers introduced a multi-agent framework designed to generate dialogue datasets incorporating multiple conversational skills.

Further advancing this field, in [6] in-context learning techniques are used to leverage a limited collection of expert-authored conversations as examples to synthesize social conversation datasets. In [30] the researchers applied similar in-context learning approaches to generate simulated dialogues specifically optimized for task-oriented dialogue systems.

In contrast, our work is the first to apply knowledge distillation to the generation of conversational recommendation datasets. Knowledge distillation is a technique that transfers the knowledge and generative capabilities of a large, high-performing teacher model to a smaller, lightweight student model [19]. Crucially, this process allows us to avoid direct reliance on high-resource models during inference. Once trained, the student model is capable of generating high-quality dialogues independently, thereby enabling scalable and cost-effective dataset generation.

3 Dataset Construction

To generate high-quality synthetic dialogues that reflect the diversity of real-world user behaviors in CRSs, we design DISTILLREC DIAL to reflect a range of interaction styles by constructing prompts and dialogues tailored to different user types. This section

describes how we incorporate user stereotypes into the dataset design, how we simulate realistic dialogue contexts for each, and how we generate a scalable, high-fidelity corpus using a distillation-based approach. A visual structure of our pipeline is reported in Figure 1

3.1 Data Sources

To ensure consistency and realistic dialogues, DISTILLREC DIAL is built using the Amazon Reviews dataset [20], which spans 33 item categories and includes 54.51 million users, 48.19 million items, and 571.54 million ratings. For this work, we focus on the *Movies and TV* category, comprising 6.5 million users, 747,000 items, and 17.3 million ratings. While we use this specific domain for demonstration, the proposed pipeline is domain- and dataset-independent, requiring only the availability of item features to support the generation of realistic dialogues. To improve data quality and coverage, we apply several filtering steps: items with missing or overly short names, as well as reviews associated with non-existent items, are removed, eliminating approximately 42% of items and 50% of reviews. We then apply a 12-core filtering strategy, retaining only users and items with at least 12 interactions, ensuring rich user profiles suitable for multi-item dialogue generation. The resulting dataset contains 15,597 items, 23,456 users, and 651,649 ratings.

Upon manual inspection of the item-related information, we observe that most metadata concerns physical product attributes (e.g., box dimensions), rather than content about the movies themselves. To ensure high-quality data and enable meaningful conversations, we enrich item metadata with additional information sourced from The Movie Database (TMDB)³. This includes genres, keywords, actors, directors, and overviews.

Finally, for each item, we retrieve its associated poster from the Amazon dataset and generate visual descriptions using a Large Multimodal Model, specifically Qwen2.5-VL-32B-Instruct⁴ [3]. The model is prompted to describe the overall tone of each poster, adding an additional layer of semantic context to support visually grounded dialogue generation.

³<https://developer.themoviedb.org/docs/getting-started>

⁴<https://huggingface.co/Qwen/Qwen2.5-VL-32B-Instruct>

3.2 User Stereotypes

A central design goal of DISTILLREC DIAL is to simulate interactions that are representative of how real users behave in CRS settings. To achieve this, we adopt and operationalize a set of user stereotypes inspired by prior literature in user modeling [43, 46]. These stereotypes guide the generation of user behaviors such as preference expression, initiative-taking, and goal articulation.

Crucially, our simulations are grounded in real-world recommendation data, using the Amazon Reviews dataset as the source of user preferences. This grounding ensures that the generated dialogues are not only behaviorally diverse, but also contextually rich and anchored in realistic recommendation scenarios.

Building upon this foundation, we define user stereotypes along two key dimensions: *Preference Expression* and *Intention Clarity*.

Preferences refer to a user's inherent, long-term tastes and can be categorized into three types:

- **None:** The user has no established preferences (e.g., cold-start scenario).
- **Implicit:** The user expresses preferences inferred from their historical interactions.
- **Explicit:** The user directly communicates preferences for specific item attributes.

Intentions relate to a user's immediate requirements for recommendations and fall into three categories:

- **None:** The user lacks a defined goal and engages in exploratory interaction to discover potential interests.
- **Vague:** The user has partial awareness of their needs but cannot articulate specific requirements.
- **Specific:** The user expresses clear demands that the system must prioritize.

By combining these, we arrive at five practical user stereotypes.

Stereotype Descriptions.

- (1) **Curious Newcomer** (*Preference: None, Intention: Vague*): A user with no interaction history and no well-defined goals. They are often indecisive, engaging in exploratory browsing, e.g. I'm in the mood for something fun or mysterious, open to suggestions!.
- (2) **Focused Newcomer** (*Preference: None, Intention: Specific*): A cold-start user who lacks history but knows exactly what they want. Despite being new, they have concrete preferences for the current session, e.g. I want a thriller with a strong female lead and a twist ending..
- (3) **History-Based Browser** (*Preference: Implicit, Intention: None*): A returning user with a known history, but no current intent. They rely on the system to surface items that align with past behavior, e.g. What do you think I'd enjoy next?.
- (4) **Guided Explorer** (*Preference: Implicit, Intention: Vague*): This user has a watch history but is in search of something new. They may express a desire for contrast or variety, often in broad terms, e.g. I usually watch crime dramas, but maybe something lighter this time?.
- (5) **Preference-Driven Searcher** (*Preference: Implicit and Explicit, Intention: Specific*): The most expressive user type. They not only have a history, but can clearly articulate detailed preferences—often down to specific actors, genres,

or themes, e.g. I love dark comedies with sarcastic humor. Any with Bill Murray?.

These stereotypes provide a comprehensive framework representing the spectrum of user behaviors in conversational recommendation systems. Our framework focuses on five representative user stereotypes while deliberately excluding certain theoretical combinations. We omitted combinations where users only have explicit preferences based on the assumption that users capable of articulating explicit preferences would typically reference items they previously enjoyed, thus demonstrating implicit preferences. Similarly, combinations where users express explicit preferences without clear intentions were excluded as users who can articulate specific preferences generally demonstrate some degree of intention clarity.

3.3 Prompt Design by User Stereotype

Building on predefined user stereotypes, we construct prompt templates (one per stereotype) that reflect varying degrees of user knowledge, preferences, and intent. These stereotypes act as user archetypes, enabling the simulation of diverse and realistic interaction patterns within CRSs. Each stereotype is associated with a specific configuration of accessible information, which guides both the prompt structure and the inclusion of contextual features.

For prompt generation, we first randomly assign each user to one of the predefined stereotypes, ensuring a uniform distribution across the dataset. Next, for each user, we select a target item - that is, the item that will ultimately be recommended and accepted during the generated conversation - to be included in the prompt. This item is chosen as the most recent one from the user's interaction history.

For each user, we instantiate a final prompt comprising three main components: (i) a textual description of the assigned stereotype, (ii) the target item along with relevant information conditioned on the stereotype, and (iii) when applicable, the user's interaction history, consisting of the 10 most recent items (excluding the target), enriched with any supporting metadata required by the prompt configuration.

To simulate differences in domain knowledge and familiarity with the catalogue, we vary the amount and type of information provided to different user stereotypes. Some users receive only minimal context (e.g., just the target item), while others are given broader access to their interaction history or richer metadata. This structured variation allows us to model each user type's behavior and knowledge level more realistically. Table 2 summarizes the feature inclusion across the five user stereotypes, where *target only* refers to features derived solely from the selected target item, and *history only* refers to features aggregated over the user's past interactions. An example of a complete prompt corresponding to each stereotype is available in our repository⁵.

3.4 Large-To-Small Pipeline

Having defined the user stereotypes and designed prompts tailored to simulate their behaviors, the next step is to generate multi-turn dialogues that populate the DISTILLREC DIAL dataset. To ensure both scalability and quality, we adopt a large-to-small distillation

⁵<https://github.com/swapUniba/DistillRecDial>

Table 2: Feature inclusion by User Stereotype. *Target only* refers to features derived solely from the ground truth target item of the dialogue; *History only* refers to features aggregated over the user’s past interactions.

Feature / User Stereotype	Curious Newcomer	Focused Newcomer	History-Based Browser	Guided Explorer	Preference-Driven Searcher
Genres	Target only	Target only	History only	Both	Both
Keywords	Target only	Target only	History only	Both	Both
Actors	No	Target only	No	No	Both
Directors	No	Target only	No	No	Both
Overview	No	Target only	No	No	Both
Poster description	No	Target only	No	No	Both
Liked/disliked items	No	No	Yes	Yes	Yes
Written reviews	No	No	No	No	Yes

pipeline, which allows us to first leverage a bigger LLM to produce high-quality conversations and then transfer this knowledge to a more lightweight LLM. In this section, we describe the methodology used to construct DISTILLREC DIAL. The large-to-small pipeline can be summarized in the following steps:

- **High-Quality Dialogues Generation:** A bigger LLM is used to generate high-quality conversations on the distillation subset, serving as ground truth for downstream training of the lightweight LLM.
- **Knowledge Transfer:** We perform knowledge distillation, transferring generation capabilities from the bigger teacher LLM to the lightweight student model, and aligning the student’s output to match the teacher’s style and quality.
- **Full Dataset Generation:** The fine-tuned student model is then used to generate conversations across the entire dataset, resulting in the final DISTILLREC DIAL corpus.

Notably, this pipeline is generalizable and can be applied across all Amazon dataset categories.

High-Quality Dialogues Generation. From the entire set of dataset generation prompts, we randomly selected a 10% distillation subset that we used to prompt the teacher LLM (i.e. LLaMA 3.3 70B⁶ [2]) to generate high-quality, human-like recommendation dialogues. These conversations are conditioned on the formatted prompts and aim to simulate natural interactions between a recommender system and a user. Each generated dialogue is multi-turn, as the teacher LLM is instructed to generate 5-10 turns, and reflects realistic conversational flow, showcasing preferences, disinterests, and item comparisons. These synthetic dialogues serve as the ground truth for training the smaller student model in the next stage.

Knowledge Transfer. In this stage, we perform knowledge distillation, a well-established model compression technique in which a large, high-performing “teacher” model transfers its capabilities to a smaller, more efficient “student” model [14, 19]. Traditionally, knowledge distillation involves aligning the student model’s logits or intermediate representations with those of the teacher [23, 40]. However, in our case, we adopt an offline knowledge distillation approach tailored to text generation. Specifically, we fine-tune a lightweight LLM (i.e. LLaMA 3.1 8B⁷) to mimic the output style, structure, and quality of the teacher model, using only the synthetic dialogues generated in the previous stage as supervised training

data, focusing solely on aligning the generated text [41]. By training the student to produce conversations that closely resemble those of the teacher, we effectively transfer its dialogue generation capabilities. The outcome of this phase is a refined student model capable of generating high-quality dialogues efficiently.

Full Dataset Generation. Following the student model’s supervised training, we use it to generate dialogues for the entire dataset of 23,456 user prompts. Each prompt is passed through the model to produce a multi-turn conversation, reflecting the conversational style learned from the teacher. Post-generation, we analyze and filter the outputs, removing conversations that mention movies not present in the corresponding user profile, addressing potential hallucinations by the model. After this filtering process, we obtain 21,039 dialogues. An example of a student-generated conversation per scenario is shown in Table 3.

4 Evaluation

To assess the quality, diversity, and practical utility of DISTILLREC DIAL for conversational recommender systems (CRSs), we conduct a comprehensive evaluation guided by the following research questions:

- **RQ1:** Can DISTILLREC DIAL achieve higher perceived dialogue quality than existing CRS datasets, according to human judgments across key conversational dimensions (e.g., user control, relevance, and naturalness)?
- **RQ2:** Does our user modeling framework lead to greater diversity in generated dialogues compared to existing synthetic datasets?
- **RQ3:** How well do standard CRS models perform when trained and evaluated on DISTILLREC DIAL across recommendation, conversation, and combined conversational recommendation tasks?

Our evaluation is structured around these questions and consists of three components: (1) **human evaluation** comparing dialogue quality between DISTILLREC DIAL and existing datasets (RQ1), (2) **automated analysis** of utterance similarity to measure dialogue diversity (RQ2), and (3) **benchmarking** of CRS models within the CRSLab framework [48] to assess task performance (RQ3). For comparison, we consider three representative CRS datasets: ReDial, INSPIRED, and PEARL. ReDial [29] and INSPIRED [16]

⁶<https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct>

⁷<https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

Table 3: Partial dialogue snippets from DISTILLRECDIAL, demonstrating various user stereotypes in the context of movie recommendations.

User Stereotype	Dialogue
Curious Newcomer	User: "I'm in the mood for something that's going to make me feel uneasy, you know? Something that's a little dark and maybe even a bit scary. I don't know, maybe something with a twist?" Assistant: "That sounds like an intriguing request! It seems like you're looking for a movie that's going to push some boundaries and keep you on the edge of your seat." [...]
Focused Newcomer	User: "I'm in the mood for a historical war drama that focuses on a secret mission during World War II, with a strong emphasis on camaraderie and personal sacrifice. I'd love to watch something that highlights the bravery of bomber pilots and their experiences." Assistant: "I understand you're looking for a historical war drama that explores a secret mission during World War II, with a focus on the bond between comrades and the personal costs of war." [...]
History-Based Browser	User: "I think I'm looking for something that's uplifting and educational. I want to learn something new while being entertained. I've always been fascinated by stories of people who make a difference in the world." Assistant: "I was thinking of Erin Brockovich. It's a biographical drama that tells the story of a woman who fights for justice in the face of corporate negligence and environmental pollution." [...]
Guided Explorer	User: "I'm in the mood for something exciting and adventurous, maybe with a bit of action and fantasy. I've been watching a lot of dramas and thrillers lately, but I want to mix it up a bit." Assistant: "That sounds like a great way to shake things up. When you say exciting and adventurous, are you looking for something with a lot of fast-paced action, or perhaps something with a deeper, more complex storyline?" [...]
Preference-Driven Searcher	User: "Hi, I'm looking for a movie that has a similar tone to <i>Mean Girls</i> and <i>The Wedding Date</i>, with a mix of adventure, comedy, and a touch of romance. I enjoy movies that have complex characters and unexpected plot twists." Assistant: "I understand you're looking for a movie that combines adventure, comedy, and romance, with complex characters and unexpected plot twists." [...]

Table 4: Datasets Statistics

	ReDial	INSPIRED	PEARL	DistillRecDial
# of Dialogues	11,347	999	57,277	21,039
# of Utterances	206,101	35,686	532,923	208,082
Avg. turns per Conversation	18.16	35.72	9.30	9.89
Avg. Tokens per Conversation	151.07	329.90	395.17	575.20
Avg. Tokens per Utterance	8.32	9.24	42.47	58.16

are crowdsourced datasets focused on movie recommendation dialogues. PEARL [28], in contrast, is a fully synthetic dataset generated using GPT-3.5 and curated using IMDB metadata. A summary of key statistics for all four datasets is provided in Table 4.

4.1 Human Evaluation of Dialogues Quality

To address RQ1, we conduct head-to-head human evaluations comparing DISTILLRECDIAL with ReDial, INSPIRED, and PEARL using evaluation criteria from [28]: *User Control*, *Expertise*, *Specificity*, *Relevance*, *Naturalness*, and *Consistency*, as detailed in Table 5.

For each baseline dataset, we randomly sample 100 dialogues and pair them with 100 randomly sampled dialogues from DISTILLRECDIAL. A total of 10 evaluators, all volunteer PhD and MSc

Table 5: Dialogue Evaluation Criteria

Criterion	Description
User Control	Which seeker participates more actively and provides feedback to receive recommendations in dialogues?
Expertise	Which recommender seems to be more of an expert in the movie domain?
Specificity	Which seeker is better at expressing preferences that help the recommender suggest a personalized movie?
Relevance	Which dialogue provides more relevant recommendations according to the seeker's preferences?
Flow Naturalness	Which dialogue is more natural in the progression of the dialogue?
Consistency	Which dialogue is more consistent in terms of the seeker's preferences?

students in Computer Science, were recruited to evaluate all dialogue pairs. As shown in Figure 2, evaluators consistently preferred DISTILLRECDIAL across all criteria and baselines.

Table 6: Recommendation metrics for various models.

Model	Hit@1	Hit@10	Hit@50	MRR@1	MRR@10	MRR@50	NDCG@1	NDCG@10	NDCG@50
Dialogue + Recommendation									
ReDial	0	0.0075	0.0367	0	0.0013	0.0024	0	0.0027	0.0088
INSPIRED	0.0014	0.0090	0.0452	0.0014	0.0030	0.0045	0.0014	0.0043	0.0120
KBRD	0.0009	0.0104	0.0382	0.0009	0.0026	0.0036	0.0009	0.0043	0.0100
Recommendation Only									
BERT	0.0763	0.1728	0.2763	0.0763	0.1039	0.1086	0.0763	0.1202	0.1428
GRU4Rec	0	0.0013	0.0060	0	0.0001	0.0003	0	0.0004	0.0014
SASRec	0.0040	0.0195	0.0209	0.0676	0.0040	0.0077	0.0098	0.0104	0.0208

Notably, DISTILLRECDIAL outperformed both human-generated datasets (ReDial and INSPIRED) by a substantial margin across all evaluation criteria. The only exception was the naturalness criterion, where the margin was narrower - DISTILLRECDIAL was preferred over ReDial in 82.2% of cases and over INSPIRED in 72.2%. This was expected due to the synthetic nature of our dialogues. For all other criteria, the win rates were consistently higher, reinforcing the overall superiority of DISTILLRECDIAL in perceived quality.

Compared to the LLM-generated PEARL, DISTILLRECDIAL was again preferred across all criteria, though with generally smaller margins than those observed against human-generated datasets. The narrowest margin was for the user control criterion, where DISTILLRECDIAL was preferred in 52.7% of cases. We attribute this to the less consistently active and engaged user stereotypes in our dataset, which tend to produce more exploratory dialogue interactions.

4.2 Automated Evaluation of Dialogues

Diversity

To assess RQ2, we measure turn-level utterance similarity in DISTILLRECDIAL and PEARL. We encode 2,000 dialogues from each dataset using the all-MiniLM-L6-v2 Sentence Transformer [39], then compute the average pairwise cosine similarity for each dialogue turn.

As shown in Figure 3, DISTILLRECDIAL exhibits significantly lower average similarity (0.4220) compared to PEARL (0.5984), indicating higher diversity. Notably, PEARL exhibits extremely high similarity in the first three turns - reaching a similarity of 1.0 at the first turn - suggesting a high degree of repetition or templated structure early in the dialogue. To understand the cause of this extreme similarity in the initial utterance, we manually inspected the data and found that all 2,000 sampled PEARL dialogues begin with the exact same utterance: Hi there! I'm in the mood to watch a movie. Can you recommend something?. In contrast, DISTILLRECDIAL maintains more moderate and consistent similarity across turns, reflecting the varied user stereotypes represented in the dataset.

4.3 Evaluation on Recommendation, Conversation, and Conversational Recommendation

To facilitate the adoption of DISTILLRECDIAL and ensure reproducibility, we convert the dataset into a format compatible with the

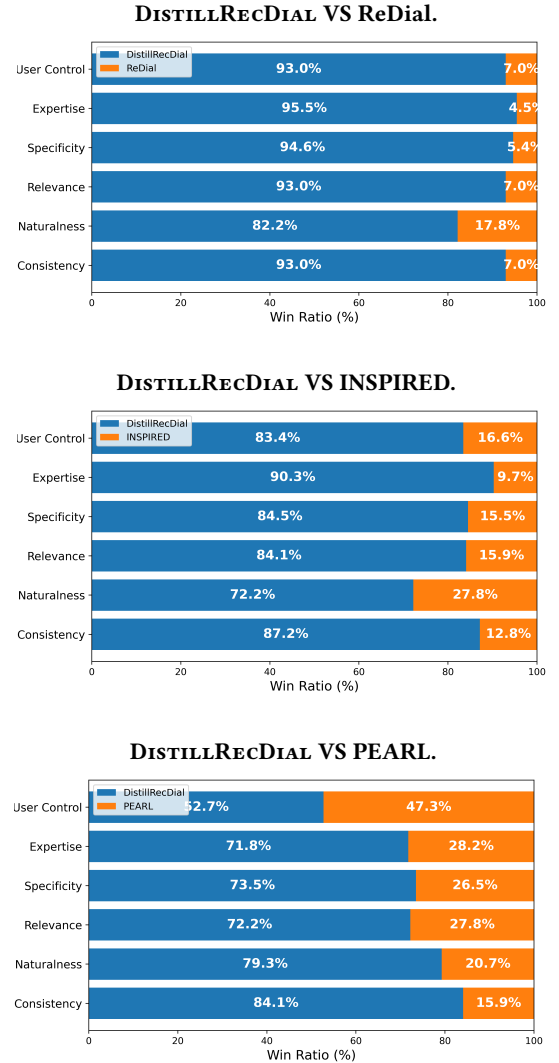
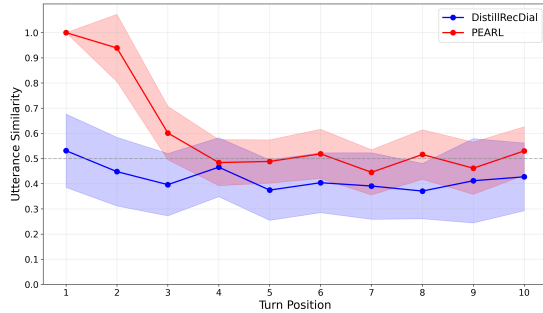
Figure 2: Results of head-to-head human comparison between dialogues sampled from DISTILLRECDIAL and existing dataset.

Table 7: Conversational (dialogue) metrics for various models. "-" indicates metric not reported.

Model	BLEU@1	BLEU@2	BLEU@3	BLEU@4	Dist@1	Dist@2	Dist@3	Dist@4	F1	Perplexity
Dialogue + Recommendation										
ReDial	0.2232	0.0576	0.0322	0.0218	0.0071	0.0596	0.2415	0.5373	0.2232	204.7
INSPIRED	0.7370	0.5788	0.4669	0.3825	0.0049	0.0760	0.4485	1.498	0.7495	2.198
KBRD	0.2986	0.1131	0.0601	0.0386	0.0072	0.1014	0.5029	1.281	0.3277	–
Dialogue Only										
GPT2	0.7297	0.5659	0.4519	0.3672	0.0060	0.0789	0.4675	1.604	0.7421	5.136

**Figure 3: Average pairwise cosine similarity of utterances at each dialogue turn for PEARL and DISTILLRECDIAL.**

CRSLab framework [48]. We then evaluate the performance of models available within CRSLab on DISTILLRECDIAL across the three CRS tasks it supports: recommendation, conversation, and conversational recommendation, which combines both aspects to improve response quality and recommendation relevance. Specifically, we benchmark several representative models on DISTILLRECDIAL to assess how well they generalize to user-aware, high-quality synthetic dialogues:

- ReDial [29]: A CRS model with a dialogue model based on HRED [42], and a recommendation model based on autoencoder. Introduced alongside the ReDial dataset.
- INSPIRED [16]: A CRS model with a dialogue model based on GPT-2 [38], and a recommendation model based on BERT [11]. Introduced alongside the INSPIRED dataset.
- KBRD [8]: A CRS model with a dialogue model based on Transformer [44], and a recommendation model based on self-attention [32].
- BERT: A recommendation model based on BERT [11]. Recommendations are generated by applying a classification head to the "[CLS]" token.
- GRU4Rec [18]: A model based on Gated Recurrent Units [9].
- SASRec [25]: A model based on Transformer [44].
- GPT-2: A dialogue model based on GPT-2 [38].

Recommendation Performance. In the recommendation task, BERT achieved the best results (Hit@10 = 0.1728, NDCG@10 = 0.1202), outperforming classical sequential recommenders such as GRU4Rec and SASRec by a large margin. This demonstrates that DISTILLRECDIAL supports nuanced recommendation behavior benefiting from pre-trained language models.

Dialogue Performance. For dialogue generation, GPT-2 and INSPIRED both produced high-quality responses (BLEU@4 = 0.3672

and 0.3825, respectively), with high diversity (Dist@4 = 1.604 and 1.498) and low perplexity (5.136 and 2.198), indicating fluency and variability. ReDial and KBRD performed significantly worse in both metrics, suggesting limited generalization to the richer dialogue structures in DISTILLRECDIAL.

CRS Models. In the conversational recommendation task, integrated models such as KBRD and INSPIRED underperformed relative to their task-specific counterparts. For example, INSPIRED yielded Hit@10 and MRR@10, substantially lower than BERT’s task-specific recommendation performance. This performance gap highlights the challenges of joint optimization over synthetic but user-rich dialogues, and points to opportunities for developing improved architectures tailored for multi-objective CRS training.

Overall, these results affirm that DISTILLRECDIAL supports effective benchmarking across core CRS tasks, while also exposing the limitations of existing models when confronted with high-quality, diverse, and user-aware synthetic dialogues.

5 Conclusions

In this work, we introduced DISTILLRECDIAL, a synthetic conversational recommendation dataset grounded in real-world data and designed to reflect diverse user types and interaction styles. Our experiments demonstrate that DISTILLRECDIAL offers higher dialogue quality compared to existing benchmarks, fostering a Human-AI symbiotic approach [36].

This dataset provides a new foundation for developing and evaluating CRSs that adapt to varied user behaviors. Beyond the dataset itself, our distillation pipeline and user modeling framework offer a reproducible approach for generating new CRS datasets.

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